

Predictive Assignment PS - 3

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Problem 3

To demonstrate the role of qualitative (ordinal) predictors in addition to quantitative predictors in multiple linear regression, Consider “diamonds” data set in R. It is in the ggplot2 package. Make a list of all the ordinal categorical variables. Identify the response.

- Run a linear regression of the response on the quality of cut. Write the fitted regression model.
- Test whether the expected price of diamond with premium cut is significantly different from that of the ideal cut.
- What is the expected price of a diamond of ideal cut?
- Modify the regression model in (a) by incorporating the predictor “table”. Write the fitted regression model.
- Test for the significance of “table” in predicting the price of diamond.
- Find the average estimated price of a diamond with an average table value and which is of fair cut.

```
set.seed(1234)
library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.5.2

library(MASS)

## Warning: package 'MASS' was built under R version 4.5.2

head(diamonds)

## # A tibble: 6 × 10
##   carat cut      color clarity depth table price      x      y      z
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>
## 1  0.23 Ideal      E      SI2     61.5    55   326   3.95   3.98   2.43
## 2  0.21 Premium    E      SI1     59.8    61   326   3.89   3.84   2.31
## 3  0.23 Good       E      VS1     56.9    65   327   4.05   4.07   2.31
## 4  0.29 Premium    I      VS2     62.4    58   334   4.2    4.23   2.63
## 5  0.31 Good       J      SI2     63.3    58   335   4.34   4.35   2.75
## 6  0.24 Very Good  J      VVS2     62.8    57   336   3.94   3.96   2.48
```

```

contrasts(diamonds$cut)=contr.sdif(5)
ord_mod=lm(price~cut,data=diamonds)
summary(ord_mod)

##
## Call:
## lm(formula = price ~ cut, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4258  -2741  -1494   1360  15348
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4062.24      25.40  159.923 < 2e-16 ***
## cut2-1       -429.89     113.85   -3.776  0.00016 ***
## cut3-2         52.90      67.10    0.788  0.43055
## cut4-3        602.50      49.39   12.198 < 2e-16 ***
## cut5-4       -1126.72     43.22  -26.067 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3964 on 53935 degrees of freedom
## Multiple R-squared:  0.01286,    Adjusted R-squared:  0.01279
## F-statistic: 175.7 on 4 and 53935 DF,  p-value: < 2.2e-16

summary(lm(price~cut,data=diamonds))

##
## Call:
## lm(formula = price ~ cut, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4258  -2741  -1494   1360  15348
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4062.24      25.40  159.923 < 2e-16 ***
## cut2-1       -429.89     113.85   -3.776  0.00016 ***
## cut3-2         52.90      67.10    0.788  0.43055
## cut4-3        602.50      49.39   12.198 < 2e-16 ***
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## Multiple R-squared:  0.01286,    Adjusted R-squared:  0.01279
## F-statistic: 175.7 on 4 and 53935 DF,  p-value: < 2.2e-16

summary(lm(price~table,data=diamonds))

```

```
##
## Call:
## lm(formula = price ~ table, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6522  -2751  -1490   1368  15746
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -9109.047    438.450  -20.78  <2e-16 ***
## table        226.984      7.625   29.77  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3957 on 53938 degrees of freedom
## Multiple R-squared:  0.01616,    Adjusted R-squared:  0.01614
## F-statistic: 886.1 on 1 and 53938 DF,  p-value: < 2.2e-16
```

```
aggregate(price~cut,data=diamonds,mean)
```

```
##      cut      price
## 1    Fair 4358.758
## 2    Good 3928.864
## 3 Very Good 3981.760
## 4   Premium 4584.258
## 5    Ideal 3457.542
```

```
head(diamonds$cut)
```

```
## [1] Ideal      Premium    Good        Premium    Good        Very Good
## attr(,"contrasts")
##           2-1  3-2  4-3  5-4
## Fair      -0.8 -0.6 -0.4 -0.2
## Good       0.2 -0.6 -0.4 -0.2
## Very Good  0.2  0.4 -0.4 -0.2
## Premium    0.2  0.4  0.6 -0.2
## Ideal      0.2  0.4  0.6  0.8
## Levels: Fair < Good < Very Good < Premium < Ideal
```

```
lvl=as.numeric(diamonds$cut)
```

```
code_fun=function(x){
  if(x-5==0){
    return(c(0,0,0,0))
  }else if(x-5==-1){
    return(c(1,0,0,0))
  }else if(x-5==-2){
    return(c(1,1,0,0))
  }else if(x-5==-3){
    return(c(1,1,1,0))
  }else{

```

```

    return(c(1,1,1,1))
  }
}
coded_pred=t(sapply(lvl,code_fun))
colnames(coded_pred)=c("Premium","Very_Good","Good","Fair")
diamonds=cbind(diamonds,coded_pred)
ord_mod=lm(price~Premium+Very_Good+Good+Fair,data=diamonds)
summary(ord_mod)

##
## Call:
## lm(formula = price ~ Premium + Very_Good + Good + Fair, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4258  -2741  -1494   1360  15348
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3457.54     27.00 128.051 < 2e-16 ***
## Premium       1126.72     43.22  26.067 < 2e-16 ***
## Very_Good     -602.50     49.39 -12.198 < 2e-16 ***
## Good          -52.90     67.10  -0.788  0.43055
## Fair           429.89    113.85   3.776  0.00016 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3964 on 53935 degrees of freedom
## Multiple R-squared:  0.01286,    Adjusted R-squared:  0.01279
## F-statistic: 175.7 on 4 and 53935 DF,  p-value: < 2.2e-16

```